Multilevel Deformable Attention-Aggregated Networks for Change Detection in Bitemporal Remote Sensing Imagery

Xiaokang Zhang, *Member, IEEE*, Weikang Yu, and Man-On Pun, *Senior Member, IEEE*

***Abstract*— Deep learning (DL) approaches based on convolutional encoder–decoder networks have shown promising results in bitemporal change detection. However, their performance is limited by insufficient contextual information aggregation because they cannot fully capture the implicit contextual dependency relationships among feature maps at different levels. Moreover, harvesting long-range contextual information typically incurs high computational complexity. To circumvent these challenges, we propose multilevel deformable attention-aggregated networks (MLDANets) to effectively learn long-range dependencies across multiple levels of bitemporal convolutional features for multiscale context aggregation. Specifically, a multilevel change-aware deformable attention (MCDA) module consisting of linear projections with learnable parameters is built based on multihead self-attention (SA) with a deformable sampling strategy. It is applied in the skip connections of an encoder–decoder network taking a bitemporal deep feature hypersequence (BDFH) as input. MCDA can progressively address a set of informative sampling locations in multilevel feature maps for each query element in the BDFH. Simultaneously, MCDA learns to characterize beneficial information from different spatial and feature subspaces of BDFH using multiple attention heads for change perception. As a result, contextual dependencies across multiple levels of bitemporal feature maps can be adaptively aggregated via attention weights to generate multilevel discriminative change-aware representations. Experiments on very-high-resolution (VHR) datasets verify that MLDANets outperform state-of-the-art change detection approaches with dramatically faster training convergence and high computational efficiency.**

***Index Terms*— Change detection, deep learning (DL), multihead self-attention (SA), multiscale context, remote sensing.**

I. Introduction

CHANGE detection from multitemporal remote sensing imagery has long been intensively investigated in the field of remote sensing for the monitoring of land resources, natural hazards, and ecosystems [1], [2]. Various change detection methodologies have been developed in the literature, while more sophisticated techniques are emerging [3]. Generally speaking, change detection methods based on bitemporal imagery can be categorized into methods based on image algebra, image transformation, and machine learning [4]. Algebraic approaches utilize image differencing or change vector analysis (CVA) to generate difference features containing the change magnitudes [5]. In contrast, image transformation methods, such as the principal component analysis [6] and histogram trend similarity [7], attempt to enhance the available change information. Based on either difference features or spatiotemporal features, machine-learning-based methods classify pixels into changed and unchanged classes [8], [9]. To improve the robustness and discriminant power of machine learning algorithms, enormous efforts have been devoted to dimension reduction, feature extraction, and optimization [10], [11].

In recent years, driven by rapid developments in machine learning techniques, change detection methods based on deep learning (DL) techniques have attracted substantial research attention due to their superior performance in mining representative features and spatiotemporal relationships from bitemporal image data [12]. In the literature, existing DL-driven change detection methods can be roughly classified into approaches based on: 1) deep features; 2) pixel patches; and 3) encoder–decoder networks. In the following, we will briefly review the three categories.

Change detection models depend upon distinguishable feature representations of bitemporal images to identify changed areas [4]. Differing from traditional handcrafted features, deep features generated by deep neural networks (DNNs) or convolutional neural networks (CNNs) can effectively characterize change information by exploiting abstract and conceptual high-level representations of bitemporal images. Since deep features exhibit strong domain invariance capabilities, methods in the first category typically rely on pretrained CNN models adapted through transfer learning to extract deep features of bitemporal images [13], [14]. Subsequently, difference features can be generated by means of distance metrics [15], [16]. After that, the change map can be obtained by applying classification algorithms in an unsupervised manner [17], [18]. Despite the fact that these methods are simple for implementation, the deep features generated by fine-tuning the resulting models using

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transfer learning cannot fully characterize the representations of the imagery used for testing [4].

Furthermore, pixel-patch-based change detection methods use image patch pairs centered around the same location in two images as the inputs to DL networks and attempt to predict the label of each patch pair [19], [20]. Specifically, CNNs from self-supervised learning or adversarial learning usually serve as an unsupervised feature extractor [21], [22]. By combining the feature extractor with a classifier, change detection is converted into an equivalent problem of classifying the image patches into changed and unchanged classes in an end-to-end manner. To effectively derive the spatiotemporal dependencies between bitemporal images, recurrent CNN architectures have been proposed for pixel-patch-based change detection [23], [24]. Unfortunately, the performance of these pixel-patch-based methods was found sensitive to the choice of the patch size and handicapped by the information redundancy problem incurred by the features extracted from the image patches [25].

Finally, encoder–decoder-based change detection methods have been proposed to address the dense pixel classification task, i.e., a semantic segmentation problem. U-Net [26] and its variants have become the most popular choice for change detection. In general, a dual-path or Siamese convolutional network is employed as the encoder to extract bitemporal deep features and a convolutional network as the decoder to merge the deep features and generate pixelwise predictions [27]. To preserve fine-grained segmentation details, skip connections are utilized to connect the encoding and decoding streams. Currently, encoder–decoder-based methods have become one of the most developed branches of change detection, and enormous efforts have been made to improve their feature representation capabilities for change detection. For instance, stronger backbones, such as residual networks (ResNets) [28] and densely connected convolutional networks (DenseNets) [29], have been adopted to augment the feature representation in change detection networks.

Since context modeling is critical for vision tasks, various schemes have been developed for aggregating contextual information in change detection networks, such as applying multiscale blocks with pyramid pooling for incrementing the receptive field [30] and aggregating multilevel features through the convolutional block [31], nested U-Net [32], multilevel deeply supervised nets [33], or graph convolutional network [34]. However, the fusion of multilevel feature maps enables contextual information to be integrated through a homogeneous context extraction procedure without taking into account various contextual interactions among different positions [35]. To overcome this drawback due to the local convolution operators, some studies have embedded context aggregation modules based on the self-attention (SA) mechanism into change detection networks to capture full-image dependencies between any two elements in a feature map [25], [36]. However, these models neglect the contextual dependency relationships across multiple feature levels [37]. As a result, they cannot sufficiently exploit multiscale contextual information, which may have an adverse effect on the change detection performance. In addition, the extraction of long-range contextual information usually incurs high computational complexity [38], especially for very-high-resolution (VHR) remote sensing imagery.

Motivated by the aforementioned challenges, this work proposes multilevel deformable attention-aggregated networks (MLDANets) for bitemporal change detection. The proposed MLDANets employ the encoder–decoder architecture with special emphasis on aggregating multilevel contextual information for change detection in an efficient way. A multilevel change-aware deformable attention (MCDA) module is developed based on a multihead SA mechanism with a deformable sampling strategy for change-aware representation learning. The idea of deformable attentions [39] is derived from deformable convolutional networks [40], [41], in which the convolution operations are enhanced by the weighted combinations of feature values of the deformable sampling locations in accordance with learnable weight coefficients. MCDA takes a bitemporal deep feature hypersequence (BDFH) as input to exploit multilevel discriminative change-aware representations from original bitemporal feature maps.

The main contributions of this study are summarized as follows.

1) We develop a lightweight change detection network with a multi- and cross-level attention aggregation scheme to facilitate multilevel context modeling using BDFH and only one MCDA module.
2) The MLDANets identify changes by adaptively aggregating multiple attention heads, which allows the model to jointly characterize a variety of beneficial information from different spatial and feature subspaces of BDFH to generate change-aware representations.
3) A deformable sampling strategy is introduced and regularized with spatial constraints to make each attention head progressively address a set of sampling locations at multiple levels. As a result, attention heads can capture contextual dependencies across multiple levels of bitemporal feature maps to exploit multilevel discriminative representations. Furthermore, the computational cost of the SA mechanism can be considerably reduced.

This article is organized as follows. In Section II, a brief review is given regarding change representations, attention-augmented learning, and attention aggregation schemes in change detection networks. Section III describes the proposed MLDANets for change detection, while the performance of MLDANets is experimentally evaluated in Section IV. In Section V, the experimental results are thoroughly discussed with the limitations of MLDANets and several potential avenues for future research being identified. Finally, Section VI concludes this article.

II. Related Work

A. Change Representation

Learning change-aware representations is of central significance in encoder–decoder-based change detection networks. Feature difference and concatenation are two common ways of perceiving semantic changes. Feature differences have been widely used in change detection networks, and they can be

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directly used to measure change intensities and highlight change information [31], [33]. In this process, pixelwise operations (e.g., L1 and L2 distance) are constructed to enhance change-aware representation, and noises are inevitably introduced, such as pseudochanges and omissions, especially when high-dimensional representations extracted from bitemporal imagery have extremely diverse feature spaces. Despite the fact that bitemporal feature concatenation can naturally retain temporal dependency [32], [42], feature fusion using the convolutional functions [42] cannot fully exploit the spatiotemporal relationships of bitemporal features and may even lead to vanishing gradient and insufficient learning problems during network training [4]. Therefore, more robust change-aware representations are needed for encoder–decoder-based change detection networks. Based on the observation that the multihead attention mechanism allows the model to focus on contents from different spatial and feature subspaces, we propose to apply the multihead attention mechanism on the bitemporal features to learn different kinds of useful semantic information and identify changes by aggregating these attention heads.

B. Attention-Augmented Learning

Encoder–decoder networks are the predominant approach for change detection tasks. However, these models suffer from the inductive bias of the local receptive fields and limited context modeling in CNNs [43]. To address these limitations, the attention mechanism has enjoyed widespread adoption as an embedding module for visual tasks to recalibrate convolutional feature maps [38], [44]. More recently, SA has emerged as a new spin on attention mechanisms to model long-range dependencies [45], [46]. Notably, nonlocal (NL) blocks [43] were proposed to model full-image contextual information by exploiting the interactions between any two positions in an image, regardless of their distance. On this basis, dual-attention (DA) [47] combining position and channel attention modules was employed to help identify the informative parts of images. Global context (GC) blocks [48] combined NL blocks with the squeeze-excitation (SE) network [44] to aggregate more specific GC. Criss-cross attention (CCA) [35] economized the computational budget of SA through the criss-cross path of each pixel. Recurrent thrifty attention [38] further reduced the computational cost of global contextual information acquisition using a dynamic downsampling operation. However, these SA modules only process long-range interactions within a single-level feature map and, thus, neglect the long-range dependency relationships across different levels. In [37], NL operations within the same level and cross-level interactions were separately encoded using various SA modules, which is inflexible in practice. Furthermore, NL-based attentions incur high computational complexity in deriving massive attention maps [38].

As shown in Fig. 1(a), the NL operation on image data can be described as mapping a *query* against a set of *key-value* pairs, where the *query*, *key* and *value* are all projected features, and the output is a weighted combination of *value* in accordance with *attention* [45], [49]. Specifically, the attention

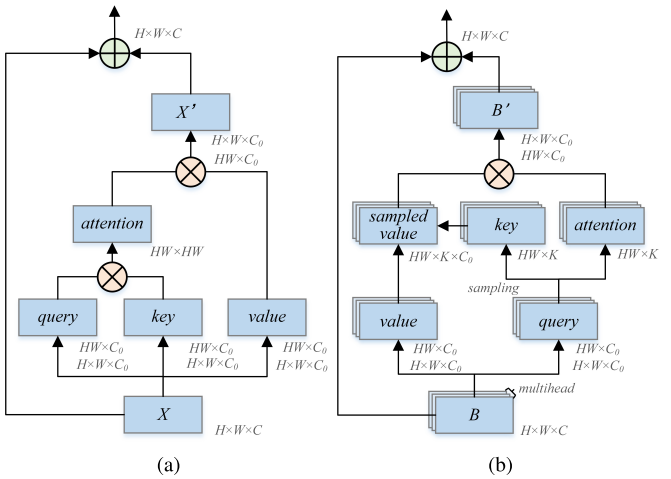

Fig. 1. Illustrations of the differences between (a) classic NL operation on image data and (b) our MCDA.

weight assigned to each value is computed based on the compatibility of the corresponding *query–key* pair. For the feature map denoted by X with height H and width W, the overall SA map has a size of $(HW) \times (HW)$, which means that, for each query position in the image, there is a corresponding subattention map whose size is HW [43], [47]. In comparison, multilevel feature maps are transformed into the BDFH $\mathcal{B}$, as the input of MCDA and the *key* in MCDA that derives from the *query* using the deformable sampling strategy addresses specific sampling locations for each query position. Moreover, the corresponding *attention* for the sampled values can also be computed solely from the embedded query features without the necessity of computing query–key compatibility. As a result, the overall attention map at each level has a size of $(HW) \times K$, where K refers to the number of sampling locations and $K \ll H \times W$, as illustrated in Fig. 1(b). Therefore, the computational complexity of attention maps can be considerably reduced. In contrast to NL blocks, the adopted multihead mechanism enables MCDA to jointly characterize beneficial information from different subspaces.

C. Attention Aggregation Schemes

When the SA mechanism is applied to change detection networks, a common design, namely, the single-level scheme, is to use the attention module to refine high-dimension deep feature maps in the most informative level produced by the CNN encoder [35], [36], [38], as shown in Fig. 2(a). After that, the reweighed feature maps are fed into the decoder. An alternative solution, namely, the multilevel scheme, implements multiple attention modules that are embedded in the U-Net architecture to refine the fusion of bitemporal features at different levels and filter the representations propagated through skip connections [50], [51], as displayed in Fig. 2(b). In this article, we propose multi- and cross-level schemes to facilitate context aggregation. Specifically, multilevel deep feature maps are combined to collectively capture the long-range contextual dependencies within the same level and across multiple levels using only one attention module. After

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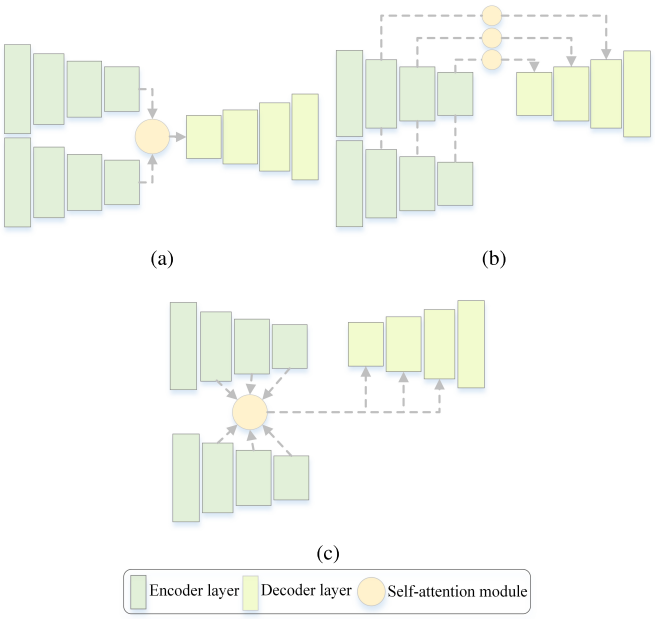

Fig. 2. Illustrations of different attention aggregation schemes in change detection networks. (a) Single-level. (b) Multilevel. (c) Multi- and cross-level.

the attention aggregation, the encoded representations are separately passed to the corresponding levels of the encoder through skip connections, as shown in Fig. 2(c).

III. MLDANet

This section provides a detailed description of the proposed change detection approach based on MLDANets, as shown in Fig. 3. First, multilevel convolutional feature maps are extracted from the bitemporal images through the Siamese CNN encoder. After that, a BDFH containing multiscale information is constructed through feature transformation. Subsequently, the BDFH is fed into the MCDA module to aggregate long-range contextual information and learn change-aware representations. Finally, the encoded representations at multiple levels are separately passed to the corresponding levels of the CNN decoder before pixelwise predictions are generated.

A. Siamese CNN Encoder

In this study, we use a Siamese CNN encoder to obtain a series of hierarchical convolutional features from bitemporal images pixel by pixel. The Siamese CNN encoder with shared parameters can generate bitemporal deep features in the same feature space that is directly comparable. As a result, deep feature maps at multiple levels of abstraction and resolutions can be extracted by the Siamese encoder.

Let $\{f_1^l\}_{l=1}^L$ and $\{f_2^l\}_{l=1}^L$ be the deep feature maps at L levels for the prechange and postchange images, respectively. The lth level feature map is of height H^l and width W^l. Instead of using the feature differences between bitemporal deep feature maps for change detection, we combine the bitemporal feature maps $\{\delta^l\}_{l=1}^L$ via channelwise concatenation to preserve the spatiotemporal association between the bitemporal images, i.e., $\delta^l = [f_1^l; f_2^l]$.

B. Feature Transformation

On the basis of a multihead SA mechanism, we establish the MCDA module from the perspective of sequence-to-sequence encoding for extracting long-range contexts. To this end, the bitemporal feature maps $\{\delta^l\}_{l=1}^L$ at multiple levels are transformed by a 1×1 convolution into the same feature dimensions C and, subsequently, flattened into sequences $\{\hat{\delta}^l\}_{l=1}^L$ as follows:

$$\hat{\delta}^l = \text{Flatten}\big(\text{Conv}_{1\times 1}\big(\delta^l\big)\big) \tag{1}$$

where $\hat{\delta}^l \in \mathbb{R}^{H^l W^l \times C}$ is the feature sequence at the lth level. The resulting multilevel feature sequences are then stacked to form a BDFH $\mathcal{B}$ providing multiscale information, i.e., $\mathcal{B} = [\hat{\delta}^1; \hat{\delta}^2; \ldots; \hat{\delta}^L]$, where $\mathcal{B} \in \mathbb{R}^{\sum_l (H^l W^l) \times C}$.

C. MCDA Module

The MCDA module takes a BDFH as input and outputs multilevel change-aware feature representations. As illustrated in Fig. 4, the proposed MCDA contains four linear projections with learnable parameters to model the deformable sampling, the projection of feature values, the learning of attention weights, and multihead values aggregation, respectively.

1) Reference Frame: To standardize the locations of pixels in multilevel feature maps, we establish a reference frame using an abstract Cartesian coordinate system and a set of reference points. As shown in Fig. 5(a), the reference frame can divide the 2-D image spaces into grids of cells. Specifically, the pixel centers in the lth level feature map are indexed by the spatial coordinates $[\ell_x^l, \ell_y^l]$ along the x- and y-axes, where $\ell_x^l \in \{0.5, 1.5, \ldots, W^l - 0.5\}$ and $\ell_y^l \in \{0.5, 1.5, \ldots, H^l - 0.5\}$. For the zth element belonging to the lth level in the BDFH, the normalized reference location $\boldsymbol{p}_z$ can be obtained as follows:

$$\boldsymbol{p}_z = \left[\frac{\ell_x^l}{W^l}, \frac{\ell_y^l}{H^l}\right]. \tag{2}$$

2) Positional Encoding: To inject spatial information into the BDFH and ensure that each query feature has a unique positional vector, we supplement each query element in BDFH with a spatial positional encoding based on its 2-D reference location [52] before passing BDFH into the MCDA module. More specifically, sine and cosine functions are independently applied to obtain positional encodings PE^l at the lth level along the x- and y-axes as follows:

$$\text{PE}_\rho^l = \begin{cases} \text{PE}\big(\hat{\ell}_\rho^l, 2j\big) = \sin\left(\dfrac{\hat{\ell}_\rho^l}{t^{2j/(C/2)}}\right) \\ \text{PE}\big(\hat{\ell}_\rho^l, 2j+1\big) = \cos\left(\dfrac{\hat{\ell}_\rho^l}{t^{2j/(C/2)}}\right) \end{cases} \tag{3}$$

where ρ denotes x or y, $t = 10\,000$ is a constant, and $j \in [0, \ldots, C/4)$ indexes the channel dimension of the features. For a BDFH with C dimensions, sine and cosine functions

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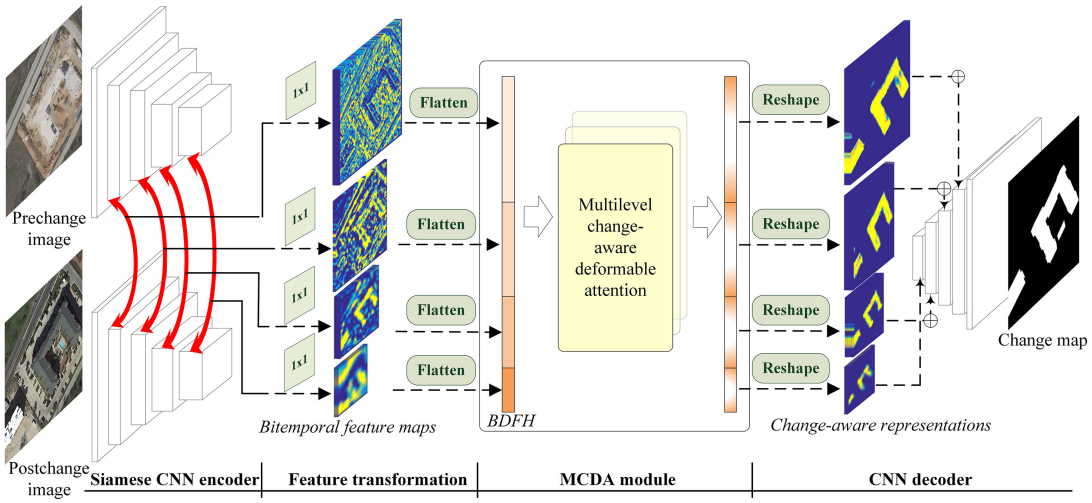

Fig. 3. Proposed MLDANet framework for bitemporal change detection.

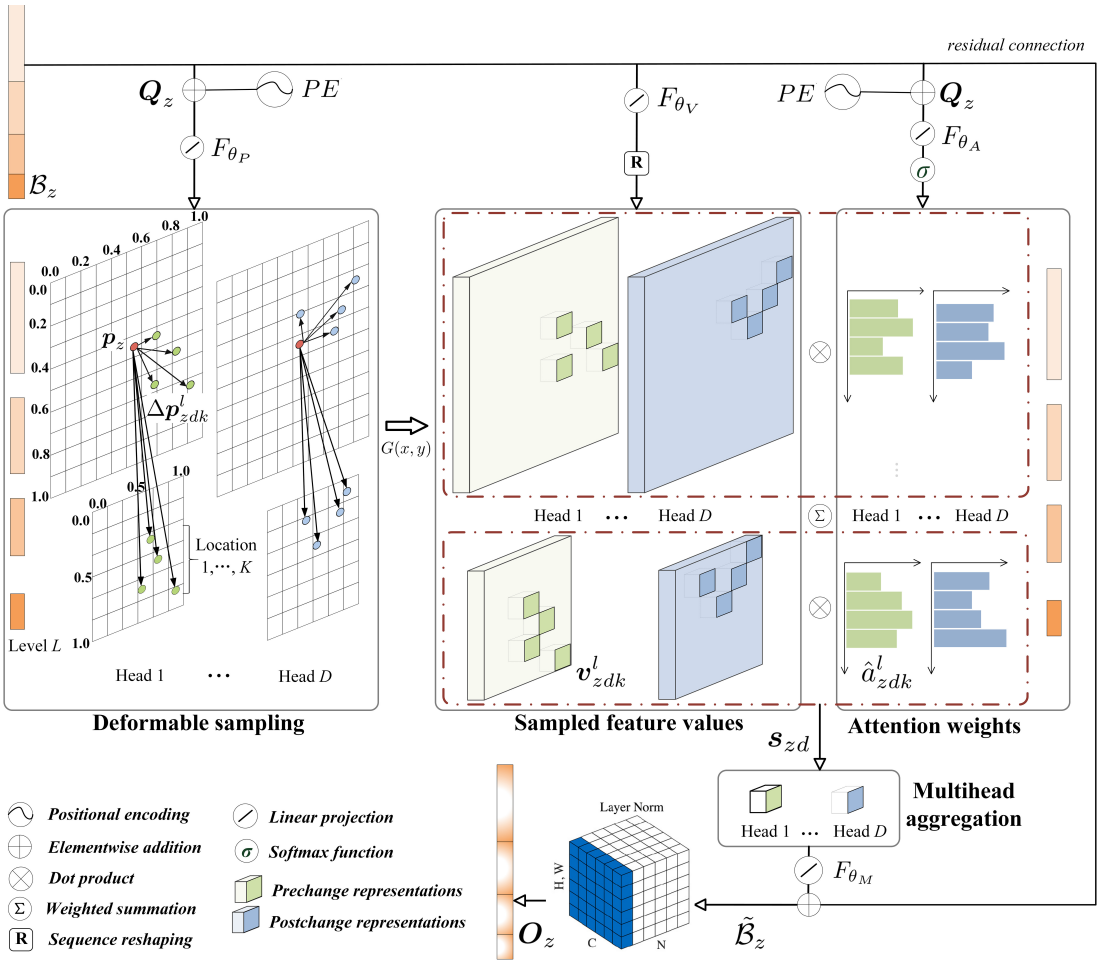

Fig. 4. Schematic illustration of the MCDA module. F_{θ_P}, F_{θ_V}, F_{θ_A}, and F_{θ_M} are linear projections with learnable parameters.

with different frequencies each correspond to encodings with $C/2$ dimensions. Note that the spatial coordinates of each pixel are normalized to the range $(0, 2\pi)$, i.e., $\hat{\ell}_x^l = 2\pi \ell_x^l / W^l$ and $\hat{\ell}_y^l = 2\pi \ell_y^l / H^l$. Therefore, the sine and cosine curves can be completed within one cycle, which ensures that there are no duplicated values for each position with different dimensions j. As a result, the positional encodings fall within the range $[-1, 1]$.

The positional encodings PE_x^l and PE_y^l along the x- and y-axes are then concatenated to obtain the final C-channel

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encodings PE^l, i.e., $\mathrm{PE}^l = [\mathrm{PE}^l_x; \mathrm{PE}^l_y]$, with the same dimensions as the input feature sequence $\hat{\delta}^l$, i.e., $\mathrm{PE}^l \in \mathbb{R}^{H^l W^l \times C}$. After that, the query features $\boldsymbol{Q}$ can be obtained through the elementwise summation of the features in the BDFH $\mathcal{B}$ and the corresponding positional encodings $\boldsymbol{PE}$, i.e., $\boldsymbol{Q} = \mathcal{B} \oplus \boldsymbol{PE}$.

3) Deformable Sampling: Based on the multihead attention mechanism, the MCDA module learns to separate prechange and postchange features into different subspaces using multiple attention heads for change-aware representation learning. For D attention heads in the MCDA and a BDFH with C dimensions, each single attention head focuses on contents from its own representation subspace with $C_0 = C/D$ dimensions.

Inspired by deformable convolutional networks [40] and deformable transformers [39], the deformable sampling strategy is introduced to make the MCDA module progressively focus on a specific set of key spatial sampling locations containing informative features according to the reference location of each query element, as illustrated in Fig. 4. For each query element in the BDFH, we wish to obtain K sampling locations at L levels for D attention heads. Therefore, each sampling set contains DLK sampling locations, which can be obtained by calculating the relative offsets with respect to the location of each query element.

Given the query feature $\boldsymbol{Q}_z \in \mathbb{R}^{1\times C}$ of the zth element in the BDFH, the sampling offsets $\Delta \boldsymbol{p}_z$ are learned through a linear mapping F_{θ_P} with learnable parameters $\theta_P = \{\boldsymbol{W}_P, \boldsymbol{b}_P\}$ as follows:

$$\Delta \boldsymbol{p}_z = F_{\theta_P}(\boldsymbol{Q}_z) = \boldsymbol{W}_P \boldsymbol{Q}_z^{\mathrm{T}} + \boldsymbol{b}_P \tag{4}$$

where $\boldsymbol{W}_P \in \mathbb{R}^{2DLK\times C}$ denotes the weight matrix for the computation of the sampling offsets and $\boldsymbol{b}_P \in \mathbb{R}^{2DLK\times 1}$ is the bias vector.

To constrain the sampling locations and obtain sparse attention patterns, a specially designed bias is utilized to regularize the learning of $\Delta \boldsymbol{p}_z$. More specifically, given a circle centered at the location of the query element with a radius of k where $k \in \{1, 2, \ldots, K\}$, we utilize D vectors with different directions to divide the circle into D equal parts, as illustrated in Fig. 5(b). Then, we can find a circumscribed square of the circle to locate the sampling points. Note that the magnitudes of the vectors are normalized to the range $[k, \sqrt{2}k]$ with the coordinates of their terminal points given by

$$\tilde{\boldsymbol{b}}_{dk} = \left[\frac{k\cos(2\pi(d-1)/D)}{\phi_d}, \frac{k\sin(2\pi(d-1)/D)}{\phi_d}\right] \tag{5}$$

where $\tilde{\boldsymbol{b}}_{dk}$ is the bias vector of the kth point for the dth head where $d \in \{1, 2, \ldots, D\}$ and $\phi_d = \max\{|\cos(2\pi(d-1)/D)|, |\sin(2\pi(d-1)/D)|\}$ is a scaling factor.

We utilize this regularization strategy at all levels to obtain the $\boldsymbol{b}_P$ of length $2DLK$, i.e., $\boldsymbol{b}_P = [\tilde{\boldsymbol{b}}_{11}, \ldots, \tilde{\boldsymbol{b}}_{DK}]^{\mathrm{T}}$, which does not participate in gradient calculations or backpropagation in the network training procedure. Given the 2D reference location $\boldsymbol{p}_z$ of the query element and its sampling offsets $\Delta \boldsymbol{p}_z$, the sampled location can be obtained by $\boldsymbol{p}_z + \Delta \boldsymbol{p}^l_{zdk}$.

4) Sampled Feature Values: Given the feature $\mathcal{B}_z \in \mathbb{R}^{1\times C}$ of the query element z in the BDFH $\mathcal{B}$, the projected feature values $\boldsymbol{V}_z$ can be obtained using a linear transformation F_{θ_V} with learnable parameters $\theta_V = \{\boldsymbol{W}_V, \boldsymbol{b}_V\}$ as follows:

$$\boldsymbol{V}_z = F_{\theta_V}(\mathcal{B}_z) = \boldsymbol{W}_V \mathcal{B}_z^{\mathrm{T}} + \boldsymbol{b}_V \tag{6}$$

where $\boldsymbol{W}_V \in \mathbb{R}^{C\times C}$ is the projection matrix of the feature values and $\boldsymbol{b}_V \in \mathbb{R}^{C\times 1}$ is the bias vector.

The projected feature values $\boldsymbol{V}$ are then reshaped into the feature maps $\boldsymbol{V}'$ that correspond to the 2-D reference locations $\boldsymbol{p}$. Similar to deformable convolutional networks [40], bilinear interpolation is applied to compute the sampled feature values, as illustrated in Fig. 5(c). The bilinear interpolation function G can be expressed by the following formula:

$$\begin{aligned} G(x, y) \approx\ & (1-\langle x\rangle)(1-\langle y\rangle)\boldsymbol{V}'(\lfloor x\rfloor, \lfloor y\rfloor) \\ & + \langle x\rangle(1-\langle y\rangle)\boldsymbol{V}'(\lfloor x\rfloor + 1, \lfloor y\rfloor) \\ & + (1-\langle x\rangle)\langle y\rangle\boldsymbol{V}'(\lfloor x\rfloor, \lfloor y\rfloor + 1) \\ & + \langle x\rangle\langle y\rangle\boldsymbol{V}'(\lfloor x\rfloor + 1, \lfloor y\rfloor + 1) \end{aligned} \tag{7}$$

where $\lfloor . \rfloor$ denotes the floor function, $\langle x\rangle := x - \lfloor x\rfloor$ is the fractional part of x, and $\langle y\rangle := y - \lfloor y\rfloor$.

Then, the sampled feature values can be obtained through the bilinear interpolation as follows:

$$\boldsymbol{v}^l_{zdk} = G\left(\left[\boldsymbol{p}_z + \Delta \boldsymbol{p}^l_{zdk}\right]_x, \left[\boldsymbol{p}_z + \Delta \boldsymbol{p}^l_{zdk}\right]_y\right) \tag{8}$$

where $\boldsymbol{v}^l_{zdk} \in \mathbb{R}^{C_0\times 1}$ refers to the sampled feature values of the kth location for the dth head at the lth level. Finally, we can obtain DLK sampled feature values with C_0 channels for each query element.

5) Attention Weights: An attention mechanism uses a compatibility function to learn alignment scores between a query and corresponding keys. Various forms of alignment score functions, such as cosine similarity, dot products, and scaled dot products [45], [53], [54], have been explored. Here, we use a location-based function in which the alignment scores

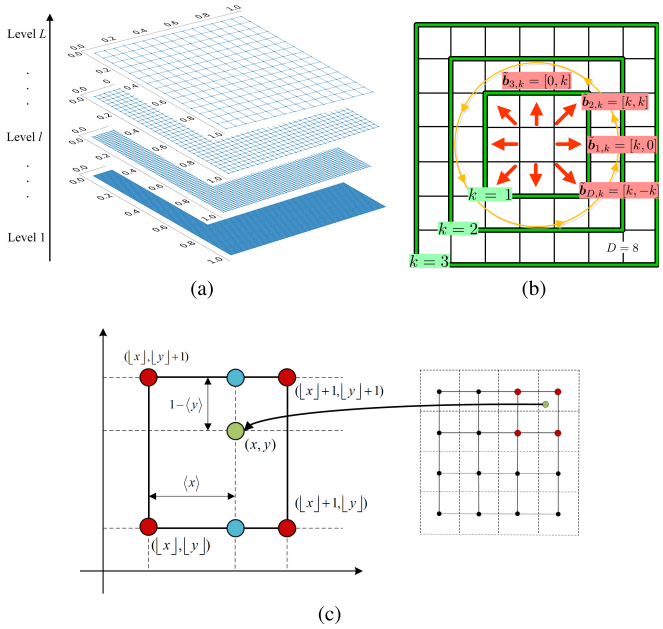

Fig. 5. Detailed illustrations of the MCDA module. (a) Reference frame. (b) Regularization of the sampling locations. (c) Bilinear interpolation.

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are computed solely from the query features with positional encodings [54]. This simplifies the alignment score function as it depends only on the query features with positional embeddings obtained via linear projection, as displayed in Fig. 4. The alignment scores are then normalized by a softmax function to model the output probability. The formulas are given by

$$\boldsymbol{A}_z = F_{\theta_A}(\boldsymbol{Q}_z) = \boldsymbol{W}_A \boldsymbol{Q}_z^{\mathrm{T}} + \boldsymbol{b}_A \tag{9}$$

$$\hat{a}_{zdk}^l = \sigma\left(a_{zdk}^l\right) = \frac{\exp\left(a_{zdk}^l\right)}{\sum_l \sum_k \exp\left(a_{zdk}^l\right)} \tag{10}$$

where $\boldsymbol{A}_z$ refers to the alignment scores and F_{θ_A} is a linear projection with learnable parameters $\theta_A = \{\boldsymbol{W}_A, \boldsymbol{b}_A\}$, where $\mathbf{W}_A \in \mathbb{R}^{DLK \times C}$ denotes the projection matrix for the attention computation and $\boldsymbol{b}_A \in \mathbb{R}^{DLK \times 1}$ is the bias vector. The softmax function σ normalizes the alignment scores to the range [0, 1], i.e., $\sum_l \sum_k \hat{a}_{zdk}^l = 1$, where $\hat{a}_{zdk}^l$ is the normalized attention weight.

6) Value Aggregation: After we obtain the sampled feature values and the attention weights, the single-head output can be generated by aggregating these feature values in accordance with their attention weights, as illustrated in Fig. 4. Since the positional encoding information has been embedded in the sampled feature values through the learning of the sampling locations, as expressed in (4), and the attention weights derived from the query features also contain the positional encodings, as expressed in (9), the sampled feature values and the attention weights have been spatially aligned. Therefore, we can directly obtain the aggregated feature value $\boldsymbol{s}_{zd} \in \mathbb{R}^{C_0 \times 1}$ of a single attention head by taking the sum of its sampled feature values weighted by the corresponding attention weights

$$\boldsymbol{s}_{zd} = \sum_l \sum_k \hat{a}_{zdk}^l \boldsymbol{v}_{zdk}^l. \tag{11}$$

The independent head outputs are then concatenated and transformed via linear projection for the aggregation of the feature values of all attention heads as follows:

$$\begin{aligned} \tilde{\mathcal{B}}_z &= \mathcal{B}_z + F_{\theta_M}\left(\left[\boldsymbol{s}_{z,1}; \boldsymbol{s}_{z,2}; \ldots; \boldsymbol{s}_{z,D}\right]\right) \\ &= \mathcal{B}_z + \boldsymbol{W}_M\left[\boldsymbol{s}_{z,1}; \boldsymbol{s}_{z,2}; \ldots; \boldsymbol{s}_{z,D}\right] + \boldsymbol{b}_M \end{aligned} \tag{12}$$

where $\theta_M = \{\boldsymbol{W}_M, \boldsymbol{b}_M\}$ refers to learnable parameters of the linear projection F_{θ_M}, where $\boldsymbol{W}_M \in \mathbb{R}^{C \times C}$ is the projection matrix and $\boldsymbol{b}_M \in \mathbb{R}^{C \times 1}$ is the bias vector. Furthermore, residual connection is utilized to generate the multihead output $\tilde{\mathcal{B}}_z$ while retaining most of information in the BDFH.

After that, layer normalization [55] serves as a regularizer during training to generate the final output $\boldsymbol{O}_z$ of the MCDA module as follows:

$$\boldsymbol{O}_z = \gamma \cdot \frac{E\left[\tilde{\mathcal{B}}_z\right]}{\sqrt{\mathrm{Var}\left[\tilde{\mathcal{B}}_z\right] + \epsilon}} + \beta \tag{13}$$

where $E[.]$ and $\mathrm{Var}[.]$ denote the mean and variance functions, respectively. Furthermore, γ and β are learnable affine transform parameters, whereas ϵ is a constant, i.e., $\epsilon = 1e-5$.

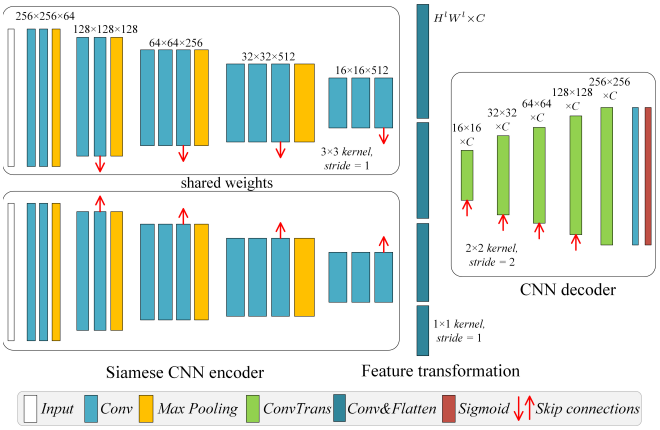

Fig. 6. Detailed architectures of the Siamese CNN encoder and the CNN decoder.

D. CNN Decoder

The output feature representations of the MCDA module are reshaped to their original size and separately passed to the corresponding levels of the CNN decoder through elementwise summation, as shown in Fig. 3. In MLDANets, a general CNN decoder is employed to recover the feature map resolution for pixelwise change detection. It progressively transforms the low-resolution feature maps to the full input resolution using multiple transposed convolutional layers.

E. Network Structures

The detailed structures of MLDANet are shown in Fig. 6. The Siamese CNN is adapted from the VGG-16 network [56], which is composed of 13 convolutional layers, five max-pooling layers, and three fully connected layers. The layers before the fifth pooling layer of VGG-16 are taken as the backbone of the Siamese CNN encoder. Therefore, the Siamese encoder contains 13 convolutional layers and four max-pooling layers. In the feature transformation, L convolutional layers with a kernel size of 1×1 followed by the flatten operations are adopted to establish multilevel feature sequences with the same channel dimension. Furthermore, the decoder consists of five transposed convolutional layers with a kernel size of 2×2 pixels and a stride of 2, a convolutional layer with a kernel size of 3×3 pixels and a stride of 1, and a sigmoid layer.

F. Loss Function

In this article, the widely used weighted binary cross-entropy loss [25], [31] is adopted to measure the difference between the predicted change map $\hat{P}_m$ and the actual label map P_m as follows:

$$\mathcal{L} = -\frac{1}{M}\sum_m^M \left[\eta_- P_m \log \hat{P}_m + \eta_+ (1 - P_m) \log\left(1 - \hat{P}_m\right)\right] \tag{14}$$

where M is the total number of samples, whereas η_+ and η_- represent the ratios of the numbers of changed and unchanged pixels, respectively, to the total number of pixels in the label map.

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IV. Experimental Results

A. Baseline Approaches

The proposed MLDANet is compared against several state-of-the-art encoder–decoder-based methods: a fully convolutional Siamese difference (SiamDiff) network [27], a feature difference convolutional neural network (FDCNN) [31], a fully convolutional network with pyramid pooling (FCNPP) [30], a dual attentive fully convolutional Siamese network (DASNet) [36], and an integrated Siamese network and nested U-Net (SNUNet) [32]. These methods also use multilevel deep features to learn useful representations for change detection based on different strategies for contextual information aggregation. Of these methods, SiamDiff is a typical Siamese-network-based U-Net model in which skip connections are utilized to concatenate multilevel difference representations from the encoding streams. FDCNN directly utilizes a feature fusion network for multilevel deep difference features. FCNPP incorporates multilevel features to capture a wider receptive field based on pyramid pooling. DASNet uses the SA mechanism in the channel and spatial dimensions to capture long-range dependencies. Finally, SNUNet was proposed to aggregate multiscale information based on densely connected convolutions.

B. Evaluation Metrics

For quantitative evaluation, five commonly used metrics are employed: the overall accuracy (OA), precision, recall, F1-score, and kappa coefficient (KC). The OA represents the percentage of changed pixels that are correctly detected among all samples. The precision is the number of correctly detected changed pixels divided by the number of all pixels identified as changed in the change detection map. The recall is the percentage of changed pixels that are correctly detected among all pixels that should be identified as changed according to the ground truth. The F1-score is the harmonic mean of the precision and recall. The KC reflects the agreement between the change detection map and the ground truth. The OA, F1-score, and KC reveal the overall change detection performance. Compared with the OA, the F1-score and KC can more objectively indicate the accuracy of the change detection results and are more useful for uneven class distributions. They are calculated according to the following equations:

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{15}$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{16}$$

$$\text{F1-score} = 2\text{TP}/(2\text{TP} + \text{FP} + \text{FN}) \tag{17}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{18}$$

$$\text{KC} = (P_o - P_e)/(1 - P_e) \tag{19}$$

where

$$P_o = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN}) \tag{20}$$

and

$$P_e = \frac{(\text{TP} + \text{FN})(\text{TP} + \text{FP}) + (\text{FP} + \text{TN})(\text{FN} + \text{TN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2}. \tag{21}$$

Furthermore, TP is the number of correctly detected changed pixels, FP is the number of false alarms, TN is the number of correctly detected unchanged pixels, and FN is the number of missed changed pixels. P_o and P_e denote the observed agreement and expected agreement, respectively. The receiver operating characteristic (ROC) curve is also employed to quantitatively evaluate the change detection maps. It is generated by plotting the true-positive rate (TPR) against the false-positive rate (FPR) with different thresholds, where $\text{TPR} = \text{TP}/(\text{TP} + \text{FN})$ and $\text{FPR} = \text{FP}/(\text{FP} + \text{TN})$, respectively.

C. Dataset Description

In this article, three change detection datasets are employed for the evaluation of the proposed MLDANet for building detection at varying scales, landslide mapping (LM), and all-object change detection in complex scenes.

1) LEVIR Dataset: The LEVIR dataset [57] contains 637 Google Earth image pairs of size 1024 × 1024 pixels with a resolution of 0.5 m. The dataset covers 20 different regions in the state of Texas in the United States of America and focuses on changes in building instances, with a total of 31 333 individual changed buildings [57]. The sensing time of the image data varies from 2002 to 2018. Seasonal differences and variations exist in the bitemporal images, making this a challenging dataset for change detection. The dataset was randomly split into three sets: 445 image pairs for training, 64 for validation, and 128 for testing. After that, each image was cropped into patches of size 256 × 256 pixels. Finally, we obtained 7120 patch pairs as training samples and 1024 pairs as the validation set in the LEVIR dataset.

2) LM Dataset: The LM dataset [21] contains bitemporal image pairs consisting of optical aerial orthophotos acquired by the Zeiss Aerial Survey Camera System in Hong Kong, China. The preevent and postevent aerial orthophotos were acquired in December 2007 and November 2014, respectively, with the same spatial resolution of 0.5 m. The changes of interest in the bitemporal images are rainfall-triggered landslides. The LM dataset covers five typical subareas located at various landslide sites where different types of landslides occur. Irregularly shaped landslide objects present challenges for change detection. The LM dataset was randomly cropped into image patches of size 256 × 256 pixels with 30% of the patches taken as training data and the rest reserved as test data through random sampling. Subsequently, the training samples were augmented by shifting and rotation. Finally, 280 patch pairs were obtained as the training samples, and 63 patch pairs were used for evaluation. We used the LM dataset to verify the effectiveness of the proposed MLDANet for small-sample learning.

3) SECOND Dataset: The SECOND dataset [58] is a large-scale semantic segmentation and change detection dataset. It contains 2968 pairs of aerial images of size 512 × 512 pixels from several platforms and sensors. These pairs of images are distributed over cities such as Hangzhou, Chengdu, and Shanghai, China. The dataset focuses on the changes in six main land cover classes, i.e., nonvegetated ground surface, trees, low vegetation, water, buildings, and playgrounds [58]. Therefore, it contains 30 common land cover

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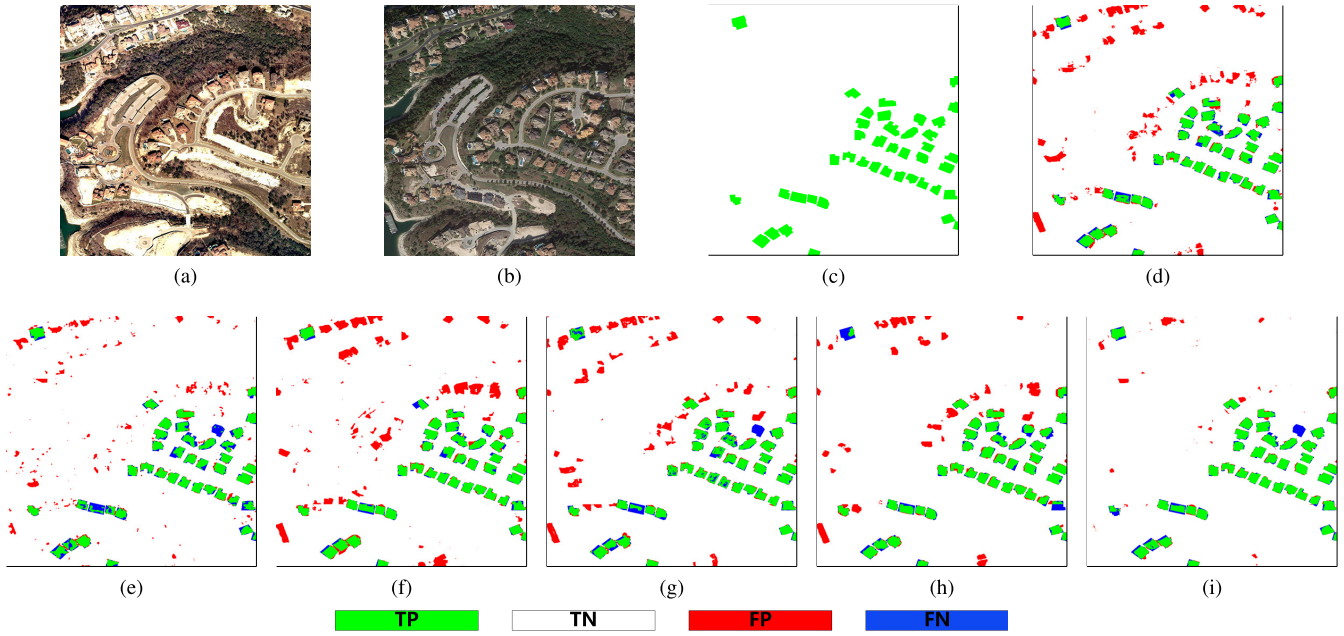

Fig. 7. Change detection results of the methods considered for comparison on the LEVIR dataset (Sample I). (a) Prechange image. (b) Postchange image. (c) Ground truth. (d)–(i) Change maps obtained by (d) SiamDiff. (e) FDCNN. (f) FCNPP. (g) DASNet. (h) SNUNet. (i) MLDANet.

change categories. Each image was cropped into 256 × 256 nonoverlapping patches. Finally, we obtained 11 872 patch pairs, 7872 of which were taken as training data, while the remaining 4000 pairs were used for evaluation.

D. Implementation Details

For the proposed MLDANet model, the stochastic gradient descent optimizer Adam [59] was chosen for network training. The parameters of Adam were set to $\beta_1 = 0$, $\beta_2 = 0.9$, and $\epsilon = 1e - 8$, as recommended in [59]. The initial learning rate was set to 0.001, and the decay by a factor of 0.5 was applied during training when the learning process stagnated. We initialized the parameters in the convolutional layers and transposed convolutional layers of the Siamese CNN encoder and the CNN decoder by applying Kaiming uniform initialization [60]. We set the number of feature levels, the number of attention heads, and the number of sampling locations at each level for each head to $L = 4$, $D = 8$, and $K = 4$, respectively. The dimension of the BDFH was set to $C = 128$ in the feature transformation. In our experiments, the parameters of the methods considered for comparison were set following the corresponding original articles. All models were implemented on the PyTorch platform. The experiments were run under the Ubuntu 18.04 Operating System with an Intel Core i9-10900X CPU and two NVIDIA GeForce RTX 2080 Ti graphics cards (11 GB of RAM).

E. Experiments on the LEVIR Dataset

1) Qualitative Evaluation: The LEVIR dataset focuses on building changes at various scales. Change detection maps for dense, large-and-sparse, and small building targets are presented to explore the effectiveness of the proposed MLDANet model for building change detection, as displayed in Figs. 7–9. It is evident that spectral variability caused by seasonal and illumination changes in the bitemporal images of the LEVIR dataset leads to areas of pseudochanges in the change detection maps of all compared methods. In the FDCNN and DASNet results, serious omissions exist in homogeneous building objects, as shown in Fig. 7(d) and (f). In comparison, the proposed MLDANet can reduce the detrimental influence of noise and preserve the internal compactness of building objects. As shown in Fig. 8, seasonal radiometric variations in land cover objects exist in the bitemporal images, such as green vegetation with different spectral features. Although FDCNN and DASNet yield somewhat fewer false alarms, edge blurring and lower internal compactness of large building objects are still introduced. In comparison, the change detection maps generated by the proposed MLDANet are closer to the ground truth. Fig. 9 displays the change detection results for small building targets. In the change detection maps of the compared methods, there are many pseudochanges introduced by spectral changes in the building roofs and different imaging conditions. In addition, many small building targets are ignored. In comparison, the proposed MLDANet accurately detects most of the small changed targets while suppressing pseudochanges because it can focus on the informative parts of features to enhance the separability between the changed and unchanged classes.

2) Quantitative Evaluation: The quantitative evaluation of the proposed MLDANet and its competitors is summarized in Table I. MLDANet achieves the best experimental performance in building change detection among all methods, with the highest OA, F1-score, and KC values. Although FCNPP achieves a slightly higher recall (0.9043) than the proposed MLDANet (0.9020), it has the lowest precision among all methods because it produces more false alarms with

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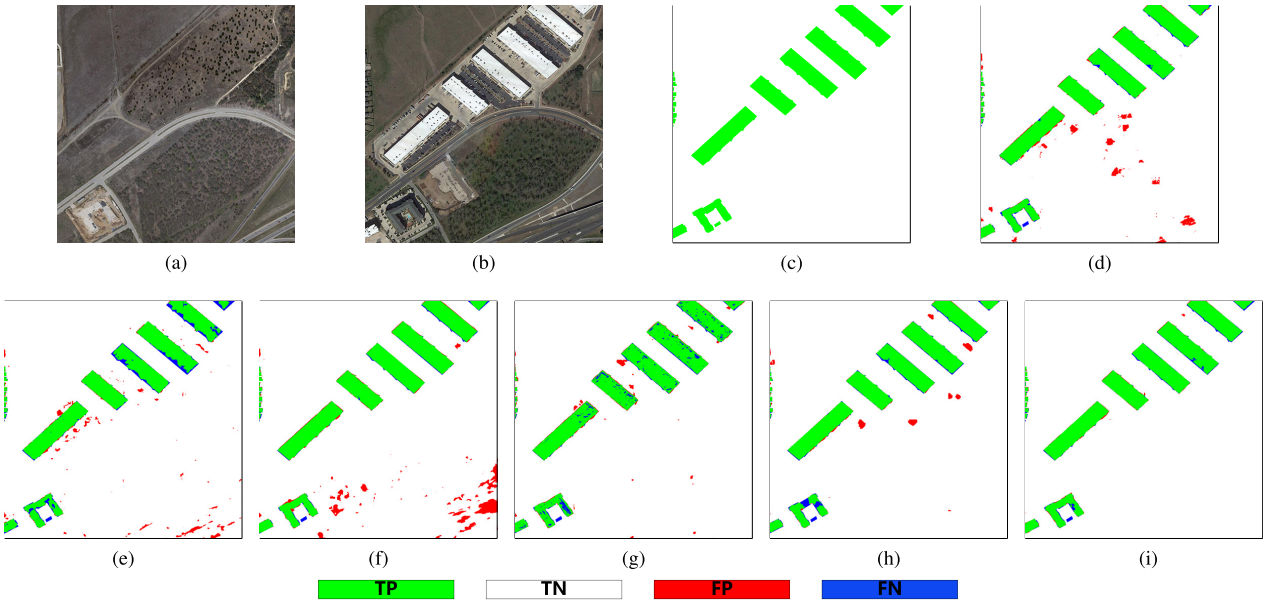

Fig. 8. Change detection results of the methods considered for comparison on the LEVIR dataset (Sample II). (a) Prechange image. (b) Postchange image. (c) Ground truth. (d)–(i) Change maps obtained by (d) SiamDiff. (e) FDCNN. (f) FCNPP. (g) DASNet. (h) SNUNet. (i) MLDANet.

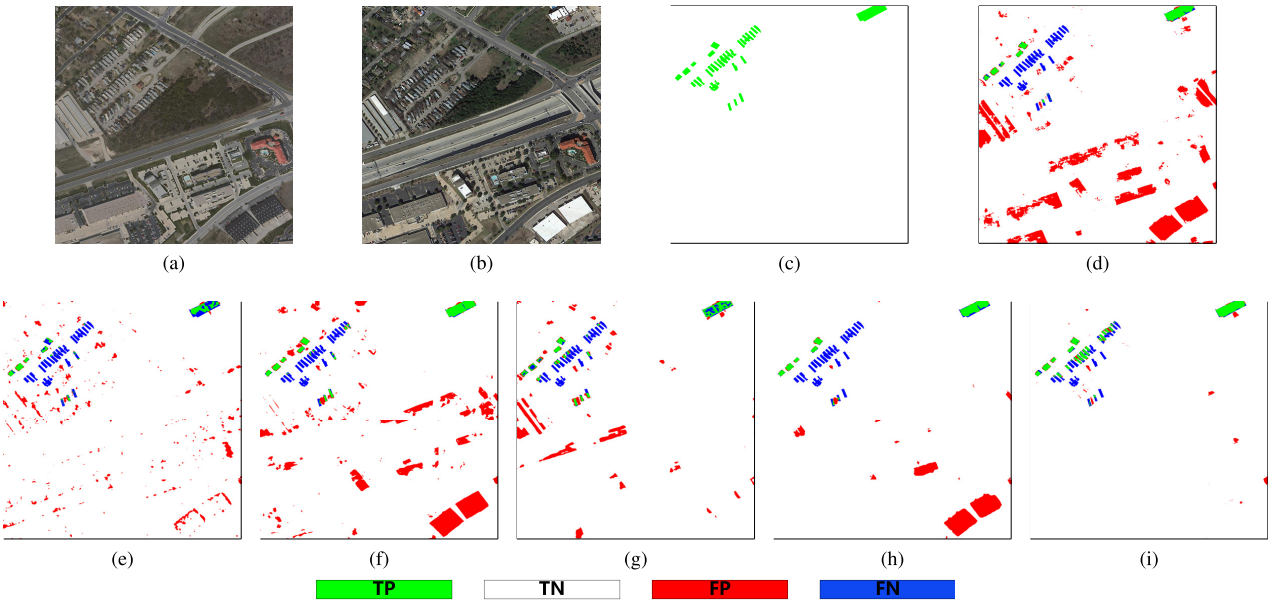

Fig. 9. Change detection results of the methods considered for comparison on the LEVIR dataset (Sample III). (a) Prechange image. (b) Postchange image. (c) Ground truth. (d)–(i) Change maps obtained by (d) SiamDiff. (e) FDCNN. (f) FCNPP. (g) DASNet. (h) SNUNet. (i) MLDANet.

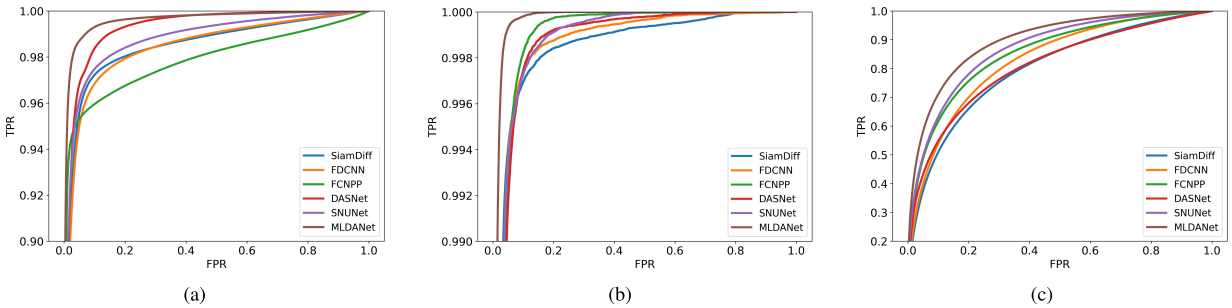

Fig. 10. ROC curves on (a) LEVIR, (b) LM, and (c) SECOND datasets.

sacrificing the performance in the unchanged areas. In comparison, MLDANet achieves a high recall while clearly excelling over the other methods in terms of precision. As shown in Fig. 10(a), the ROC curve obtained by the proposed MLDANet

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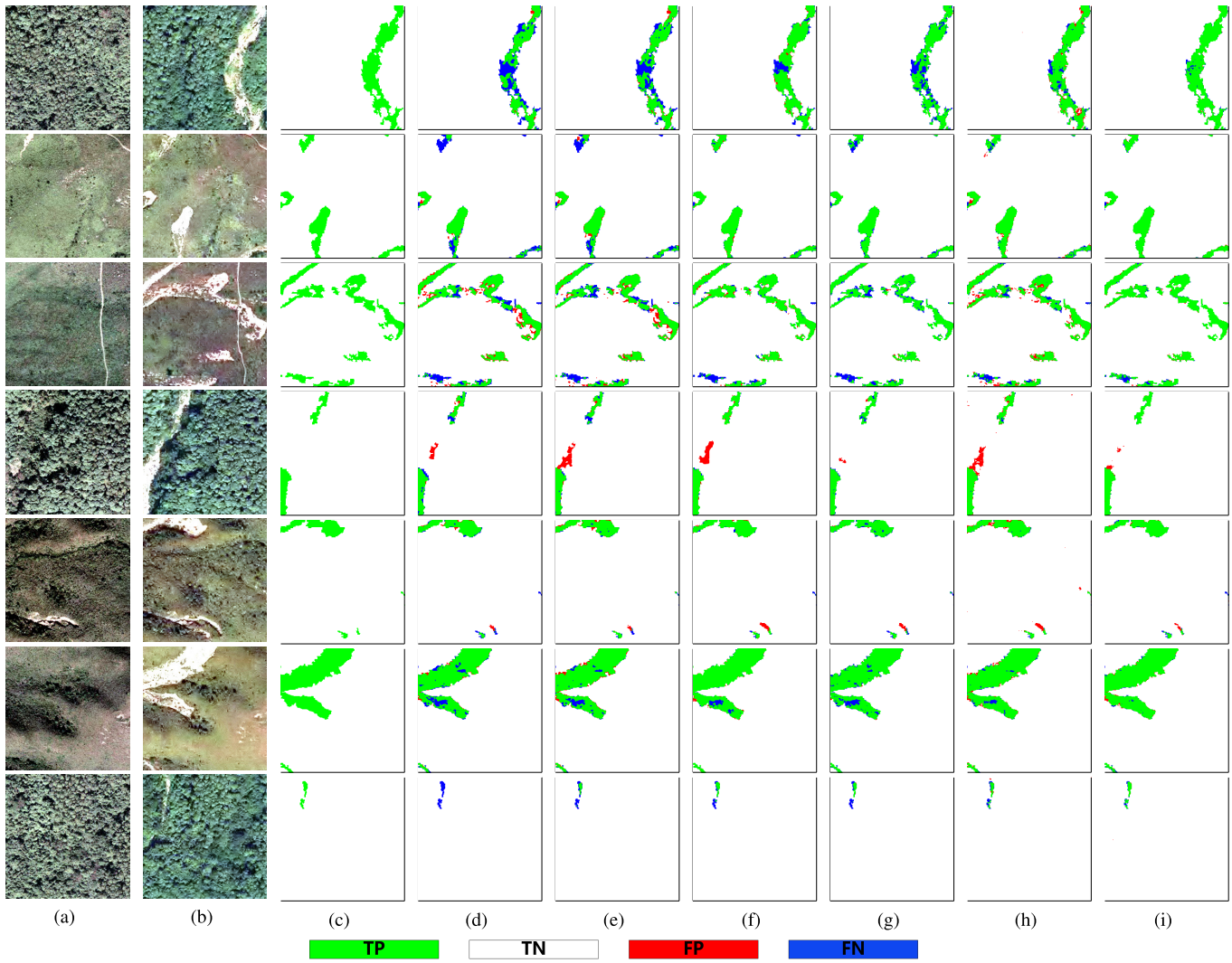

Fig. 11. Change detection results of the methods considered for comparison on the LM dataset. (a) Prechange image. (b) Postchange image. (c) Ground truth. (d)–(i) Change maps obtained by (d) SiamDiff. (e) FDCNN. (f) FCNPP. (g) DASNet. (h) SNUNet. (i) MLDANet.

TABLE I

COMPARISON OF THE ACCURACIES OF DIFFERENT CHANGE DETECTION METHODS ON THE LEVIR DATASET

Method	OA	Precision	Recall	F1-score	KC
SiamDiff	0.9829	0.7981	0.8919	0.8424	0.8333
FDCNN	0.9805	0.7754	0.8680	0.8191	0.8088
FCNPP	0.9765	0.7122	**0.9043**	0.7968	0.7845
DASNet	0.9862	0.8415	0.8978	0.8687	0.8614
SNUNet	0.9850	0.8484	0.8611	0.8547	0.8469
MLDANet	**0.9915**	**0.9308**	0.9018	**0.9157**	**0.9113**

contains points close to the top-left corner, which indicates that a higher quality change map is generated.

F. Experiments on the LM Dataset

1) Qualitative Evaluation: The LM dataset focuses on detecting landslide areas from bitemporal images through the quantification of land cover changes. As shown in Fig. 11(b), it includes different types of landslides with irregular shapes that occurred in regions of various land cover conditions with topographic heterogeneity, including dense grasslands, sparse woodlands, shrublands, and volcanic rocks. Fig. 11(c)–(h) illustrates the predictive capabilities of MLDANet and the compared methods. Salt-and-pepper noise exists in the change detection maps generated by the compared methods, which leads to false alarms and omissions, especially at the boundaries of irregularly shaped landslide objects, as shown in Fig. 11(c)–(g). In comparison, the proposed MLDANet exhibits obvious advantages over its competitors in maintaining the details of landslide objects and preserving the actual boundaries of changed objects. In addition, it is more robust to noise. Consequently, the change maps generated by MLDANet are closer to the ground truths than those of the other methods, as shown in Fig. 11(h).

2) Quantitative Evaluation: The proposed MLDANet achieves the best experimental performance among all methods on the LM dataset in terms of all evaluation indices with the highest OA, F1-score, and KC values, as shown in Table II. In addition, it generates fine-grained change maps superior to those of the other methods, as demonstrated by

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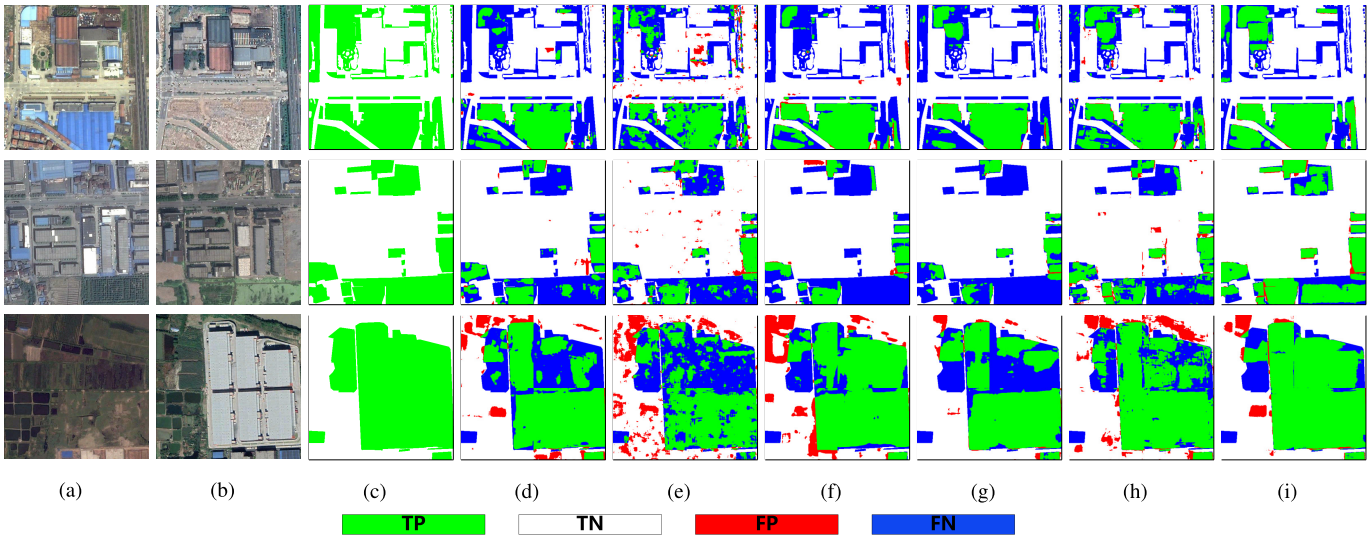

Fig. 12. Change detection results of the methods considered for comparison on the SECOND dataset. (a) Prechange image. (b) Postchange image. (c) Ground truth. (d)–(i) Change maps obtained by (d) SiamDiff. (e) FDCNN. (f) FCNPP. (g) DASNet. (h) SNUNet. (i) MLDANet.

TABLE II
COMPARISON OF THE ACCURACIES OF DIFFERENT CHANGE DETECTION METHODS ON THE LM DATASET

Method	OA	Precision	Recall	F1-score	KC
SiamDiff	0.9882	0.9273	0.8492	0.8865	0.8803
FDCNN	0.9880	0.9038	0.8882	0.8959	0.8900
FCNPP	0.9938	0.9423	0.9443	0.9433	0.9400
DASNet	0.9929	0.9526	0.8954	0.9315	0.9278
SNUNet	0.9916	0.9108	0.9379	0.9242	0.9198
MLDANet	**0.9952**	**0.9575**	**0.9538**	**0.9556**	**0.9531**

TABLE III
COMPARISON OF THE ACCURACIES OF DIFFERENT CHANGE DETECTION METHODS ON THE SECOND DATASET

Method	OA	Precision	Recall	F1-score	KC
SiamDiff	0.8574	0.7709	0.3070	0.4391	0.3743
FDCNN	0.8437	0.7050	0.4041	0.4846	0.3967
FCNPP	0.8501	0.6451	0.3894	0.4857	0.4041
DASNet	0.8630	**0.8153**	0.3185	0.4581	0.3964
SNUNet	0.8664	0.7250	0.4270	0.5374	0.4654
MLDANet	**0.8824**	0.7617	**0.5139**	**0.6137**	**0.5474**

the ROC curve illustrated in Fig 10(b), because detailed information on landslide objects can be well retained through the incorporation of multiscale contextual dependencies.

G. Experiments on the SECOND Dataset

1) Qualitative Evaluation: The SECOND dataset contains multiple land cover change categories. Thus, it is employed to evaluate the performance of the proposed MLDANet for all-object change detection in complex scenes. Fig. 12 presents the corresponding change detection maps with multiple change categories, such as bare land to buildings, bare land to trees, buildings to bare land, trees to low vegetation, and trees to buildings. All change detection maps suffer from omission errors, especially for prechange and postchange classes that have similar spectral signatures, such as trees to low vegetation, buildings to bare land, and bare land to sparse vegetation, due to the low interclass spectral variance of these land cover objects. As shown in Fig. 12(f), change detection maps obtained by DASNet exhibit fewer false negatives than those of the other methods. Nevertheless, many missed detections are also evident in the change detection maps. In comparison, although some false alarms exist, the proposed MLDANet is capable of more successfully detecting homogeneous changed objects, as shown in Fig. 12(h). In addition, it is more robust to spectral variability in VHR bitemporal images.

2) Quantitative Evaluation: From a quantitative point of view, the proposed MLDANet significantly outperforms its competitors in terms of the OA, recall, and F1-score, as shown in Table III. Although SiamDiff and DASNet achieve slightly higher precision values, more omissions within the changed objects lead to poor performance in terms of the OA and F1-score, as shown in Table III. With respect to recall, the proposed MLDANet clearly excels over the others due to its exploratory capabilities regarding exploiting changes with spectral variability. As shown in Fig. 10(c), the ROC curve of our model lies closer to the top-left corner, demonstrating the ability of MLDANet to learn representations that are more distinguishable between the changed and unchanged classes.

H. Parameter Analysis

We perform a sensitivity analysis for four important parameters, i.e., the number of attention heads, D; the number of key sampling locations at each feature level for each attention head, K; the feature dimension of the BDFH, C; and the number of feature levels in the BDFH, L.

1) Attention Heads: Multihead attention allows MLDANet to focus on different subspaces to learn diverse kinds of useful semantic information from the BDFH. In this experiment, we varied the number of heads, D, in the range from 2 to 64, while other parameters remained unchanged. As displayed

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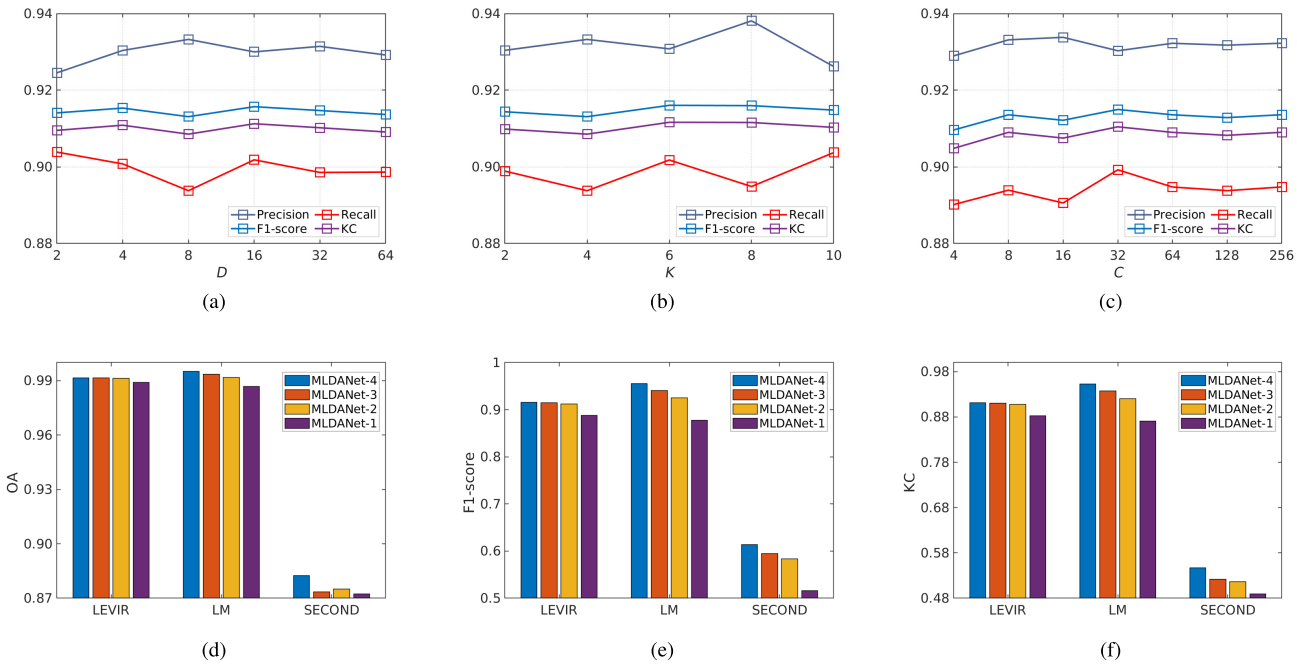

Fig. 13. (a)–(c) Results with different values of (a) D, (b) K, and (c) C on the LEVIR dataset. (d) OA. (e) F1-score. (f) KC values of MLDANet with different numbers of feature levels on the three datasets.

in Fig. 13(a), the precision and recall values fluctuate with varying D values, while the F1-score and KC remain stable at high levels. MLDANet performs best with $D = 16$. These results indicate that MLDANet is robust with respect to the number of attention heads. Multiple attention heads could be beneficial for representation learning; however, if D is too large, each single head will be unable to adequately represent its feature subspace, and the time consumption for network training will increase.

2) Sampling Locations: For each element in the BDFH, the MCDA module attempts to inspect a specific set of key sampling locations; this is imposed on the BDFH for the aggregation of prominent feature values to enhance the learning of change-aware representations. In this experiment, the number of sampling locations, K, was varied from 2 to 10 with a step size of 2. As displayed in Fig. 13(b), MLDANet is also not sensitive to K. The change detection results generated with $K = 6$ achieve the highest F1-score and KC values. Note that, when K is too large, the set of sampling locations may include adjacent locations in the feature map, possibly leading to the generation of similar feature values after bilinear interpolation. Therefore, beyond a certain threshold, increasing K does not further improve the change detection accuracy.

3) Feature Dimension: The channel dimension of bitemporal deep features was reduced to C in the feature transformation procedure to construct a BDFH with C channels. We set different C values ranging from 4 to 256. The dimensionality reduction of deep features may result in information loss in the BDFH. Therefore, MLDANet achieves lower F1-score and KC values with $C = 4$, as shown in Fig. 13(c). As C increases, the F1-score and KC values become stable at a higher level.

4) Feature Levels: We investigate the impact of the number of feature levels, L, on the change-aware representation learning capabilities of MLDANet. The BDFH contains deep features corresponding to different levels of depth and resolutions such that L fundamentally affects the aggregation of contextual information and the encoding of cross-level dependencies in the MCDA module. We used the feature map at the deepest level in the Siamese CNN encoder (with a feature size of 16×16 pixels) to implement the MCDA module with single-level deformable attention, denoted by MLDANet-1, because this yields more abstract and conceptual representations than the other levels. Correspondingly, in MLDANet-2, MLDANet-3, and MLDANet-4, we used two, three, and four levels of feature maps, respectively, with increasingly higher resolutions to implement MCDA at higher levels. In general, the multilevel networks perform better than the single-level model, and MLDANet-4 achieves the highest OA, F1-score, and KC values, as shown in Fig. 13(d)–(f). MLDANet takes advantage of multi- and cross-level dependency relationships to aggregate multiscale contextual information, thereby improving the feature representation ability. Note that, for the LEVIR and SECOND datasets, the aggregation of more levels of features in MCDA cannot progressively improve the change detection performance. Specifically, MLDANet-2 and MLDANet-3 obtain OA, F1-score, and KC values close to those of MLDANet-4 on the LEVIR dataset, as shown in Fig. 13(d)–(f). With respect to the SECOND dataset, MLDANet-2 achieves F1-score and KC values close to those of MLDANet-3, but its OA is higher than that of MLDANet-3. This indicates that, in the MCDA module, deep features at different levels may contribute to the exploration of change-aware representation to different degrees. As a result, the incorporation of some feature maps, such as the finest-resolution map (with a feature size of 128×128 pixels) for the LEVIR dataset and the higher-level map (with a feature size

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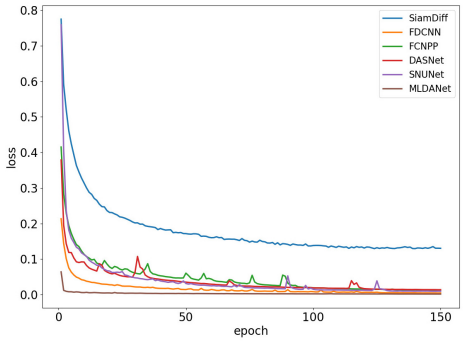
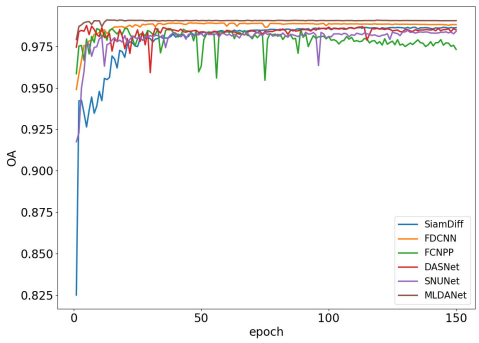


(a)



(b)

Fig. 14. Convergence performance of the proposed MLDANet versus the methods considered for comparison. (a) Training loss curves. (b) Validation OA curves.

TABLE IV
COMPUTATIONAL AND SPACE COMPLEXITY VERSUS CHANGE DETECTION PERFORMANCE

Method	Para. (M)	FLOPs (G)	Time (ms)	F1-score
SiamDiff	1.35	4.71	19	0.8547
FDCNN	1.44	39.36	15	0.8191
FCNPP	15.27	72.48	46	0.7968
DASNet	48.22	106.66	81	0.8687
SNUNet	12.03	54.82	72	0.8424
MLDANet-4	16.02	48.39	64	0.9157
MLDANet-3	15.92	42.82	51	0.9145
MLDANet-2	15.83	41.26	43	0.9124

of 64 × 64 pixels) for the SECOND dataset, may not be able to significantly improve the feature representation capabilities of the network.

I. Efficiency

Fig. 14 presents the convergence performance of the proposed MLDANet and the other compared methods during training on the LEVIR dataset. It is apparent that the proposed MLDANet converges much faster than the other networks. Specifically, the training loss decreases to a significantly low level after only two epochs, as shown in Fig. 14(a). Meanwhile, the OA on the validation dataset rapidly reaches a high level, as displayed in Fig. 14(b). This is because the deformable sampling method in the MCDA can efficiently

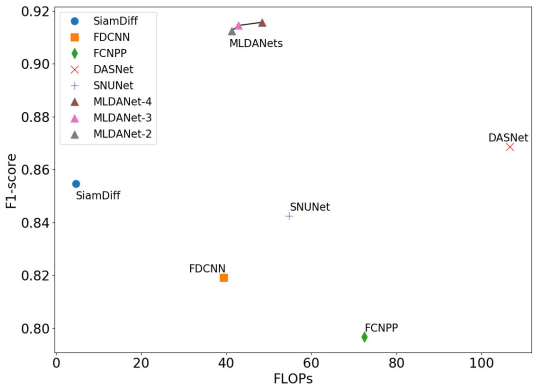


Fig. 15. Tradeoffs between FLOPs and F1-score for all change detection methods.

address the informative parts of multilevel features and boost the convergence performance of the network.

We evaluate the computational and space complexities for all networks in terms of the number of floating-point operations per second (FLOPs) and the number of trainable parameters, respectively. As shown in Table IV, the lightweight models, SiamDiff and FCNPP, have lower space complexities. The proposed MLDANets have space complexities similar to those of FCNPP and SNUNet and lower than that of DASNet. With respect to the computational complexity, MLDANet exhibits higher FLOPs than the lightweight models and lower FLOPs than FCNPP, DASNet, and SNUNet, as shown in Table IV. Compared with FCNPP, DASNet, and SNUNet that are based on pyramid pooling, dual SA, and densely connected convolutions, respectively, MLDANets rely on simple convolution blocks and flexible attention mechanism with only four linear projections to be optimized, resulting in lower computational and space complexities. Table IV also reports the inference time on GPU for a pair of input images of size 256 × 256 pixels. MLDANets achieve shorter times than DASNet and SNUNet, and in particular, the time costs of MLDANet-3 and MLDANet-2 are close to that of the FCNPP. As shown in Fig. 15, MLDANets achieve acceptable tradeoffs between FLOPs and F1-score for the change detection tasks. This demonstrates that MLDANet can obtain favorable change detection results with high computational efficiency.

J. Ablation Study

To verify the effectiveness of aggregating contextual information, the MCDA is compared with other attention modules embedded in the networks using the single-level and multilevel schemes, including DA [47], CCA [35], SE [44], NL [43], and GC [48], using the same backbone. Furthermore, we evaluate MLDANets on different backbones, including VGG-16 [56], VGG-19 [56], ResNet-50 [28], ResNet-101 [28], and DenseNet-169 [29]. As shown in Table V, multilevel-based attention modules achieve better performance than attention modules using the single-level scheme because multilevel contextual information is critical for the change detection task in high-resolution remote sensing imagery. The MCDA based on the multi- and cross-level

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TABLE V
ABLATION STUDY OF THE MLDANETS

Network	Scheme	OA	Precision	Recall	F1-score	KC	Memory (Mib)
ResNet-50+DA	single-level	0.9824	0.8236	0.8327	0.8281	0.8189	3309
ResNet-50+CCA	single-level	0.9837	0.8480	0.8276	0.8377	0.8291	3103
ResNet-50+SE	multilevel	0.9865	0.8641	0.8711	0.8676	0.8604	2965
ResNet-50+NL	multilevel	0.9880	0.8903	0.8713	0.8807	0.8743	8389
ResNet-50+GC	multilevel	0.9892	0.9299	0.8526	0.8896	0.8839	3475
ResNet-50+MCDA	multi- and cross-level	0.9912	**0.9338**	0.8896	0.9112	0.9065	1565
ResNet-101+MCDA	multi- and cross-level	0.9910	0.9331	0.8876	0.9098	0.9051	2003
DenseNet-169+MCDA	multi- and cross-level	0.9913	0.9295	0.8962	0.9126	0.9080	1513
VGG-16+MCDA	multi- and cross-level	0.9914	0.9317	0.8962	0.9136	0.9090	**1449**
VGG-19+MCDA	multi- and cross-level	**0.9914**	0.9306	**0.8981**	**0.9141**	**0.9096**	1535

schemes performs better than single-level- and multilevel-based attention modules based on the same ResNet-50 backbone, which demonstrates the superiority of the proposed MCDA for change-aware representation learning. With respect to the backbones, it is notable that VGG-19 achieves the highest F1-score and KC values among all backbones. Deeper backbones, such as ResNet-50, ResNet-101, and DenseNet-169, obtain slightly lower accuracies than VGG-16 and VGG-19 because overly abstract features may lose some essential details in these deeper networks. These findings demonstrate the robustness of MLDANets attributed to the fact that MLDANETS does not rely on stronger backbone encoders to learn discriminative representations. Therefore, we choose VGG-16 as the basic backbone in this article due to its low computational cost. We further explore the amount of memory footprint with different SA modules. As shown in Table V, MCDA based on the multi- and cross-level schemes requires almost 2× less GPU memory usage than other single-level- and multilevel-based attention modules, which demonstrates that MCDA is a computationally efficient way to capture multilevel long-range dependencies.

V. DISCUSSION

A. Multi- and Cross-Level Long-Range Dependencies

Multilevel sampling locations and the corresponding attention weights are displayed in Fig. 16. For each query position at each level of the feature map, we can obtain sampling locations in multilevel feature maps and perceive their interactions. As previously mentioned, we utilized a spatial regularization term to constrain the learning of deformable sampling with different attention heads focusing on different spatial subspaces. The resulting spatial locations are related to the coordinates of the query element and distributed over multilevel feature maps. Consequently, MLDANets can capture the long-range contextual dependencies within the same feature level and across multiple levels of feature maps.

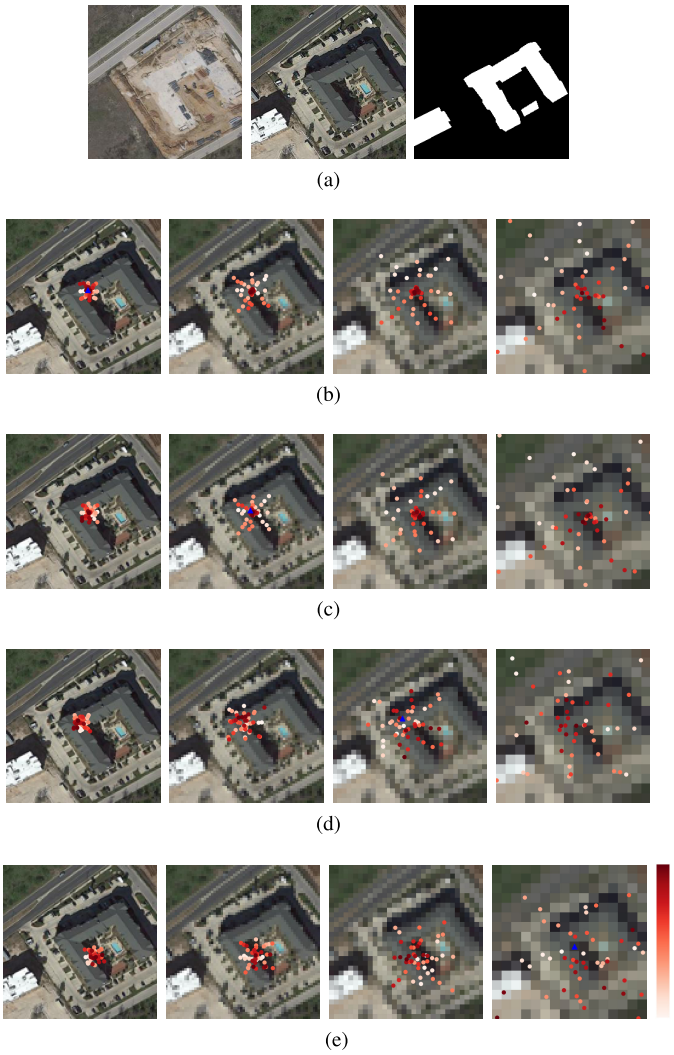

Fig. 16. Visualization of multilevel sampling locations and the corresponding attention weights. (a) Input images with ground truth. The query pixels in (b) first, (c) second, (d) third, and (e) fourth levels are shown as blue triangle markers, and their sampling locations are marked as circles, whose colors indicate corresponding attention weights. The image sizes from left to right are 128 × 128, 64 × 64, 32 × 32, and 16 × 16.

B. Learning Multilevel Change-Aware Representations

Fig. 17 illustrates the visual explanations of change detection networks with SA modules using the gradient-weighted class activation mapping (Grad-CAM) method [61]. The Grad-CAM discovers essential locations of the feature map for the final decision by tracking the gradient information flow. Consequently, class-discriminative locations in the Grad-CAM maps can show higher scores. It can be observed that the feature maps generated by the CNN encoder are not representative of the recognition of changed objects, especially for higher-resolution shallow features that contain more fine-grained

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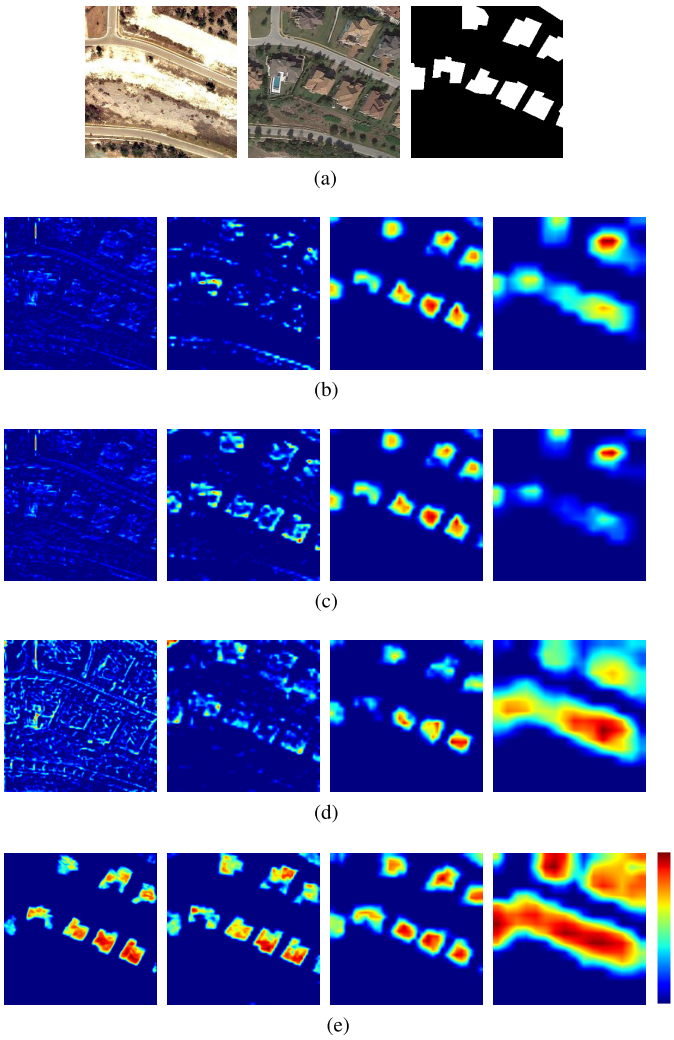

Fig. 17. Grad-CAM visualization results. (a) Input images with ground truth. (b) Multilevel maps from the encoder in SA-augmented networks. (c) Multilevel maps from SA. (d) Multilevel maps from the encoder in MLDANets. (e) Multilevel maps from MCDA. The map sizes from left to right are 128 × 128, 64 × 64, 32 × 32, and 16 × 16.

details but lose semantic information. Attention-augmented feature maps are insufficiently recalibrated by multilevel SA modules. In comparison, the MCDA-aggregated feature maps are more discriminative and change-aware at all levels due to the aggregation of multihead attention at different levels. Furthermore, it can enhance the representation and discrimination capabilities of shallow features while retaining rich local structures. Therefore, MLDANets have the ability to handle changed ground objects at different scales.

C. Limitations of the Proposed MLDANet

The main limitations of the proposed MLDANet are twofold. First, the 1 × 1 convolution operation for dimensionality reduction may result in information loss when an overly small channel dimension of BDFH is generated in the feature transformation procedure. Second, when the finest-resolution feature maps (with a feature size of 256 × 256 pixels) are incorporated for the MCDA computations, the number of trainable parameters and GPU memory consumption of the MLDANet will increase dramatically. As a result, only lower resolution feature maps are used in the experiments given the existing computational resources.

D. Future Work

In the future, we will focus on incorporating the "from-to" change sequences of the bitemporal features into the MCDA module to detect multiple types of semantic changes. Moreover, the proposed approach can be applied to other tasks, such as object extraction and remote sensing scene classification.

VI. Conclusion

In this article, we have proposed MLDANets for change detection from VHR bitemporal remote sensing imagery. The multi- and cross-level attention aggregation scheme is developed for efficiently aggregating long-range contextual information with an MCDA module embedded in the skip connections of the network structure. The MCDA is built based on the multihead SA mechanism, which consists of linear projections with learnable parameters and takes a BDFH as its input. For each query element in the BDFH, the MCDA progressively addresses a specific set of key sampling locations at multiple levels and adaptively aggregates the sampled feature values in accordance with multihead attentions with different heads encoding BDFH in different spatial and feature subspaces. As a result, MLDANet can aggregate long-range contextual dependencies among feature maps at different levels to generate multilevel change-aware representations from the original bitemporal convolutional feature maps.

Experiments on tasks including building change detection, LM, and all-object change detection in complex scenes show that MLDANet exhibits obvious advantages in change detection from bitemporal VHR datasets. Compared with state-of-the-art change detection methods, MLDANet is shown to achieve better performance with fine-grained change maps. More specifically, it can repress pseudochanges while preserving detailed changes and handling multiple types of changed objects at different scales due to its exploratory capabilities in learning multilevel discriminative change-aware representations. In addition, MLDANet achieves dramatically faster convergence performance and high computational efficiency due to the lightweight attention-aggregated network and the deformable sampling strategy, making MLDANet attractive for practical applications.

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References

[1] D. Lu, P. Mausel, E. Brondizio, and E. Moran, "Change detection techniques," *Int. J. Remote Sens.*, vol. 25, no. 12, pp. 2365–2401, 2004.

[2] M. Hussain, D. Chen, A. Cheng, H. Wei, and D. Stanley, "Change detection from remotely sensed images: From pixel-based to object-based approaches," *ISPRS J. Photogramm. Remote Sens.*, vol. 80, pp. 91–106, Jun. 2013.

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[3] L. ZhiYong, T. Liu, J. A. Benediktsson, and N. Falco, "Land cover change detection techniques: Very-high-resolution optical images: A review," *IEEE Geosci. Remote Sens. Mag.*, early access, Jul. 8, 2021, doi: 10.1109/MGRS.2021.3088865.

[4] D. Wen *et al.*, "Change detection from very-high-spatial-resolution optical remote sensing images: Methods, applications, and future directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 68–101, Dec. 2021.

[5] F. Bovolo and L. Bruzzone, "A theoretical framework for unsupervised change detection based on change vector analysis in the polar domain," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 1, pp. 218–236, Jan. 2006.

[6] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and K-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[7] Z. Y. Lv, T. F. Liu, P. Zhang, J. A. Benediktsson, T. Lei, and X. Zhang, "Novel adaptive histogram trend similarity approach for land cover change detection by using bitemporal very-high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9554–9574, Dec. 2019.

[8] F. Bovolo, L. Bruzzone, and M. Marconcini, "A novel approach to unsupervised change detection based on a semisupervised SVM and a similarity measure," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2070–2082, Jul. 2008.

[9] X. Zhang, W. Shi, P. Liang, and M. Hao, "Level set evolution with local uncertainty constraints for unsupervised change detection," *Remote Sens. Lett.*, vol. 8, no. 8, pp. 811–820, May 2017.

[10] L. Fu *et al.*, "Learning robust discriminant subspace based on joint $L_{2,p}$- and $L_{2,s}$-norm distance metrics," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 1, pp. 130–144, Jan. 2022.

[11] Q. Ye, P. Huang, Z. Zhang, Y. Zheng, L. Fu, and W. Yang, "Multiview learning with robust double-sided twin SVM," *IEEE Trans. Cybern.*, early access, Sep. 21, 2021, doi: 10.1109/TCYB.2021.3088519.

[12] W. Shi, M. Zhang, R. Zhang, S. Chen, and Z. Zhan, "Change detection based on artificial intelligence: State-of-the-art and challenges," *Remote Sens.*, vol. 12, no. 10, p. 1688, May 2020.

[13] S. Saha, F. Bovolo, and L. Bruzzone, "Unsupervised deep change vector analysis for multiple-change detection in VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3677–3693, Jun. 2019.

[14] Y. Zhan, K. Fu, M. Yan, X. Sun, H. Wang, and X. Qiu, "Change detection based on amese convolutional network for optical aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1845–1849, Apr. 2017.

[15] B. Hou, Y. Wang, and Q. Liu, "Change detection based on deep features and low rank," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2418–2422, Dec. 2017.

[16] X. Zhang, W. Shi, Z. Lv, and F. Peng, "Land cover change detection from high-resolution remote sensing imagery using multitemporal deep feature collaborative learning and a semi-supervised chan–vese model," *Remote Sens.*, vol. 11, no. 23, p. 2787, 2019.

[17] B. Du, L. Ru, C. Wu, and L. Zhang, "Unsupervised deep slow feature analysis for change detection in multi-temporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 9976–9992, Dec. 2019.

[18] T. Zhan, M. Gong, X. Jiang, and M. Zhang, "Unsupervised scale-driven change detection with deep spatial–spectral features for VHR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5653–5665, Aug. 2020.

[19] M. Yang, L. Jiao, F. Liu, B. Hou, and S. Yang, "Transferred deep learning-based change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6960–6973, Sep. 2019.

[20] M. Yang, L. Jiao, F. Liu, B. Hou, S. Yang, and M. Jian, "DPFL-nets: Deep pyramid feature learning networks for multiscale change detection," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, May 25, 2021, doi: 10.1109/TNNLS.2021.3079627.

[21] X. Zhang, M.-O. Pun, and M. Liu, "Semi-supervised multi-temporal deep representation fusion network for landslide mapping from aerial orthophotos," *Remote Sens.*, vol. 13, no. 4, p. 548, Feb. 2021.

[22] M. Gong, T. Zhan, P. Zhang, and Q. Miao, "Superpixel-based difference representation learning for change detection in multispectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2658–2673, May 2017.

[23] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral-spatial-temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[24] H. Chen, C. Wu, B. Du, L. Zhang, and L. Wang, "Change detection in multisource VHR images via deep Siamese convolutional multiple-layers recurrent neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2848–2864, Apr. 2020.

[25] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[26] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[27] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[28] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2016, pp. 770–778.

[29] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4700–4708.

[30] T. Lei, Y. Zhang, Z. Lv, S. Li, S. Liu, and A. K. Nandi, "Landslide inventory mapping from bitemporal images using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 982–986, Jun. 2019.

[31] M. Zhang and W. Shi, "A feature difference convolutional neural network-based change detection method," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7232–7246, Oct. 2020.

[32] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[33] C. Zhang *et al.*, "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[34] S. Saha, L. Mou, X. Zhu, F. Bovolo, and L. Bruzzone, "Semisupervised change detection using graph convolutional network," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 607–611, Apr. 2021.

[35] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

[36] J. Chen *et al.*, "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2020.

[37] D. Zhang, H. Zhang, J. Tang, M. Wang, X. Hua, and Q. Sun, "Feature pyramid transformer," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 323–339.

[38] L. Fu, D. Zhang, and Q. Ye, "Recurrent thrifty attention network for remote sensing scene recognition," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8257–8268, Oct. 2021.

[39] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[40] J. Dai *et al.*, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 764–773.

[41] X. Zhu, H. Hu, S. Lin, and J. Dai, "Deformable ConvNets v2: More deformable, better results," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9308–9316.

[42] B. Hou, Q. Liu, H. Wang, and Y. Wang, "From W-Net to CDGAN: Bitemporal change detection via deep learning techniques," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1790–1802, Mar. 2020.

[43] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[44] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[45] A. Vaswani *et al.*, "Attention is all you need," 2017, *arXiv:1706.03762*.

[46] S. Zheng *et al.*, "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," 2021, *arXiv:2012.15840*.

[47] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[48] Y. Cao, J. Xu, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," 2019, *arXiv:1904.11492*.

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[49] Y. Lu *et al.*, "Understanding and improving transformer from a multi-particle dynamic system point of view," 2019, *arXiv:1906.02762*.
[50] O. Oktay *et al.*, "Attention U-net: Learning where to look for the pancreas," 2018, *arXiv:1804.03999*.
[51] W. Deng, Q. Shi, and J. Li, "Attention-gate-based encoder–decoder network for automatical building extraction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 2611–2620, 2021.
[52] J. Chen *et al.*, "TransUNet: Transformers make strong encoders for medical image segmentation," 2021, *arXiv:2102.04306*.
[53] A. Graves, G. Wayne, and I. Danihelka, "Neural Turing machines," 2014, *arXiv:1410.5401*.
[54] M.-T. Luong, H. Pham, and C. D. Manning, "Effective approaches to attention-based neural machine translation," 2015, *arXiv:1508.04025*.
[55] J. Lei Ba, J. Ryan Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.
[56] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.
[57] H. Chen and Z. Shi, "A spatial-temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.
[58] K. Yang *et al.*, "Semantic change detection with asymmetric Siamese networks," 2020, *arXiv:2010.05687*.
[59] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.
[60] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1026–1034.
[61] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2017, pp. 618–626.

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